



KEMENTERIAN
PENDIDIKAN
MALAYSIA

UNIT FIZIK

Trial PSPM (DP024)

2 jam

JANGAN BUKA KERTAS SOALAN INI SEHINGGA DIBERITAHU
DO NOT OPEN THIS QUESTION PAPER UNTIL YOU ARE TO DO SO

ARAHAN KEPADA CALON:

Kertas soalan ini mengandungi **6** soalan. Jawab **semua** soalan pada buku jawapan yang disediakan.

Kalkulator elektronik boleh digunakan.

Kertas soalan ini mengandungi **8** halaman bercetak.

*This question paper consists of **8** printed pages.*

LIST OF SELECTED CONSTANT VALUES

Speed of light in vacuum	c	=	$3.00 \times 10^8 \text{ m s}^{-1}$
Permeability constant	μ_o	=	$4\pi \times 10^{-7} \text{ H m}^{-1}$
Permittivity constant	ϵ_o	=	$8.85 \times 10^{-12} \text{ F m}^{-1}$
Elementary charge	e	=	$1.60 \times 10^{-19} \text{ C}$
Planck constant	h	=	$6.63 \times 10^{-34} \text{ J s}$
Electron mass	m_e	=	$9.11 \times 10^{-31} \text{ kg}$
Neutron mass	m_n	=	$1.674 \times 10^{-27} \text{ kg}$
Proton mass	m_p	=	$1.674 \times 10^{-27} \text{ kg}$
Deuteron mass	m_d	=	$3.34 \times 10^{-27} \text{ kg}$
Universal gas constant	R	=	$8.31 \text{ J K}^{-1} \text{ mol}^{-1}$
Rydberg constant	R_H	=	$1.097 \times 10^7 \text{ m}^{-1}$
Avogadro constant	N_A	=	$6.02 \times 10^{23} \text{ mol}^{-1}$
Boltzmann constant	k	=	$1.38 \times 10^{-23} \text{ J K}^{-1}$
Gravitational constant	G	=	$6.67 \times 10^{-11} \text{ N m}^2 \text{ kg}^{-2}$
Free-fall acceleration	g	=	9.81 m s^{-2}
Atomic mass unit	$1u$	=	$1.66 \times 10^{-27} \text{ kg}$
Electron volt	eV	=	$1.6 \times 10^{-19} \text{ J}$
Constant of proportionality for Coulomb's law	$k = \frac{1}{4\pi\epsilon_o}$	=	$9.0 \times 10^9 \text{ N m}^2 \text{ C}^{-2}$
Atmospheric pressure	atm	=	$1.013 \times 10^5 \text{ Pa}$
Density of water	ρ	=	1000 kg m^{-3}

Answer **all** the questions.

- 1 (a) The equation for a mechanical wave is given as

$$y = 15 \sin(120\pi t - 5\pi x)$$

where x and y are in centimeters and t is in seconds.

- (i) State the direction of the wave propagation.
- (ii) Determine the frequency, wavelength, and speed of wave.
- (iii) Calculate the particle vibrational velocity at time, $t = 0.1$ s, for the particle at $x = 5.0$ cm.

[10 marks]

- (b) Two waves of the same frequency, amplitude and speed but traveling in opposite directions are represented by the equations,

$$y_1 = 25 \sin(150\pi t - 10\pi x)$$

$$y_2 = 25 \sin(150\pi t + 10\pi x)$$

where x and y are in centimeters and t is in seconds.

- (i) Write an equation to represent the resultant wave of the superposition of these two waves.
- (ii) Calculate the frequency and wave number.

[4 marks]

- (c) A guitar wire has a mass per unit length, $\mu = 5.16 \times 10^{-3} \text{ kg m}^{-1}$. What is the tension in the string if a wave travels along it at a speed of 434 m s^{-1} ?

[2 marks]

- 2 (a) A wire has a cross-sectional area 0.9 mm^2 and a resistance of 0.45Ω at 29°C . The wire is made of a material with a resistivity of $1.68 \times 10^{-8} \Omega \text{ m}$. When it is connected to a power source, a current of 0.5 A flows through it.

- (i) What is the length of the wire?
- (ii) How many electrons flow through the wire in 1 minute?
- (iii) Determine the voltage in the wire.

[6 marks]

(b)

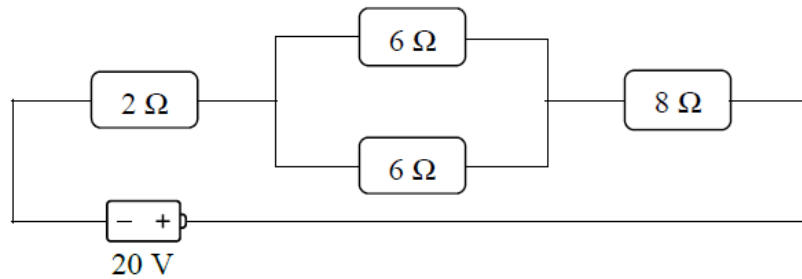


FIGURE 2

FIGURE 2 shows four resistors connected with 20 V battery. Calculate the

- (i) equivalent resistance.
- (ii) voltage across the 8Ω resistor.

[6 marks]

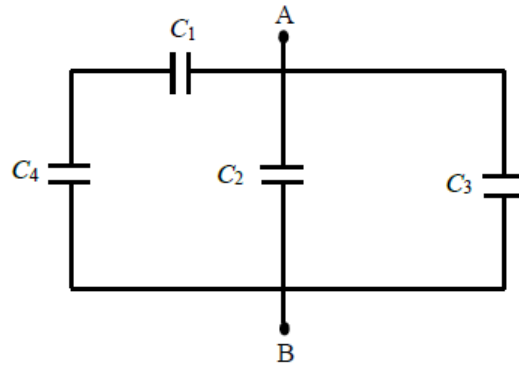


FIGURE 3a

- (a) **FIGURE 3a** shows an arrangement of four capacitors $C_1 = 15 \mu\text{F}$, $C_2 = 7 \mu\text{F}$, $C_3 = 9 \mu\text{F}$, and $C_4 = 15 \mu\text{F}$ connected between point A and B. The potential difference across points A and B is 15 V. Calculate the
- effective capacitance.
 - total charge in the circuit.

[4 marks]

- (b)

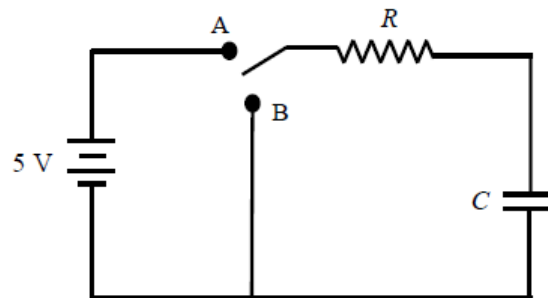


FIGURE 3b

FIGURE 3b shows a RC circuit. The capacitor has capacitance, $C = 500 \mu\text{F}$, the resistor of resistance $R = 50 \text{ k}\Omega$. The capacitor starts to charge when the switch is connected to A.

- Calculate the time constant.
- Calculate the time taken for the charge to increase to 25% of its maximum charge.
- Sketch a graph of charge against time for the charging process.

[6 marks]

4 (a)

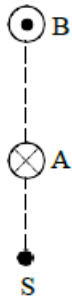


FIGURE 4a

FIGURE 4a shows two long straight wires each carrying a current of 7 A are separated by a distance of 10.0 cm. A point S is at a distance of 4.0 cm from wire A. At point S,

- (i) calculate the magnetic field caused by wire A.
- (ii) calculate the magnetic field caused by wire B.
- (iii) state the direction of each magnetic field.

[7 marks]

- (b) A rectangular coil has sides of $2.5 \text{ cm} \times 4 \text{ cm}$ is placed in a magnetic field of 1.5 T. The angle between the vector area of a coil and the magnetic field is 30° . Calculate the magnetic flux through the coil.

[2 marks]

(c)

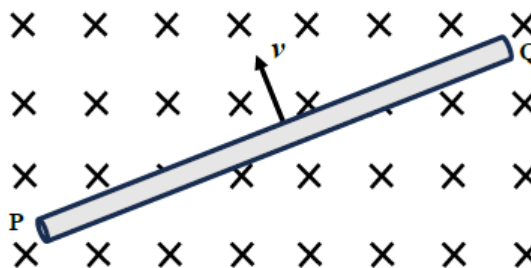


FIGURE 4b

FIGURE 4b shows a 30 cm rod PQ is moving at a velocity, v of 45 m s^{-1} across a uniform magnetic field of 0.8 T .

- (i) Calculate the induced emf
- (ii) Determine the induced current **AND** its direction if the resistance in the rod is 15Ω .

[4 marks]

- 5 (a) An object of 0.6 cm tall is placed 16.5 cm from a convex mirror having a radius of curvature of 22.0 cm.

- (i) Calculate the image distance.
- (ii) Calculate the height of the image.
- (iii) State **THREE** characteristics of the image formed by the mirror.

[7 marks]

- (b) An image of the same height as the object is formed by a convex lens.

- (i) Determine the object distance in terms of the focal length.
- (ii) Sketch a ray diagram to show the formation of the image.
- (iii) If the focal length is 8 cm and the object is moved to twice the original distance away from the lens, calculate the image distance.

[7 marks]

- 6 (a) A two-slit interference experiment in which the slits are 0.2 mm apart and the screen is 1.0 m from the slits. The third bright fringe from the central bright fringe is seen at 9.49 mm. Calculate the

- (i) wavelength of the light.
- (ii) distance between the sixth dark fringe and the seventh bright fringe from the central bright fringes will the third dark fringe occur.

[8 marks]

- (b) A screen is separated from a double-slit source by 1.20 m. The distance between the two slits is 0.03 mm and a monochromatic light of wavelength 550 nm is used.

- (i) Calculate the distance between adjacent fringes.
- (ii) Are there any changes with the fringes seen on the screen when the wavelength from the source is changed to 450 nm? Justify your answer by calculation.

[6 marks]

END OF QUESTIONS